

Long-Term Effects of COVID-19 on Memory and Cognition: A Data-Driven Analysis

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Abstract—This study investigates the long-term effects of COVID-19 on memory and cognition, with a focus on patients experiencing Long COVID. Utilizing longitudinal data, including cognitive assessments (memory, reaction time, fatigue), Principal Component Analysis (PCA) was employed to reduce data dimensionality, followed by ANOVA to identify significant differences between groups. Clustering techniques were applied to categorize patients based on the severity of cognitive symptoms. The findings reveal heterogeneity in cognitive impacts, offering insights for targeted interventions. This research contributes to a deeper understanding of the persistent cognitive deficits associated with Long COVID.

Keywords: COVID-19, Cognitive Impairment, PCA, Clustering, Regression, ANOVA, Longitudinal Data.

I. INTRODUCTION

COVID-19 has been linked to long-term neurological symptoms, including cognitive impairments such as memory loss, decreased reaction time, and fatigue. These symptoms, part of what is known as "long COVID," can persist for months or even years. Understanding these cognitive effects is essential for early diagnosis and effective treatment.

This study aims to analyze cognitive data from COVID-19 patients to identify patterns of cognitive impairment using data analysis techniques. The paper is organized as follows: Section II reviews related works, Section III describes the methodology, Section IV presents the results, Section V discusses the findings, Section VI suggests future research directions, and Section VII concludes the paper.

II. RELATED WORKS

The long-term cognitive effects of COVID-19, including memory loss, slowed reactions, and fatigue, have been widely documented. Carfi et al. (2020) showed that these symptoms persist long after infection. Sudre et al. (2021) identified predictive factors for these symptoms using data from the COVID-19 Symptom Study platform, while Mandal et al. (2021) found that fatigue and cognitive impairments are the most enduring. Methods such as Principal Component Analysis (PCA) and clustering have been used to analyze and classify cognitive data, helping to identify patterns and better target treatments (González-Iglesias et al., 2022).

III. METHODOLOGY

A. Problem Formulation

The goal of this study is to determine the cognitive effects of COVID-19 by analyzing memory scores, reaction times, and fatigue levels.

B. Dataset and Preprocessing

The dataset used is sourced from a clinical dataset containing longitudinal cognitive test results, medical history, and fatigue scores of COVID-19 patients. The preprocessing steps include:

- Data cleaning and handling missing values.
- Feature selection to retain relevant cognitive metrics.
- Normalization of continuous variables.

C. Algorithms and Methods

To analyze the dataset, we employ:

- **Principal Component Analysis (PCA)** for dimensionality reduction.
- **Clustering techniques** (e.g., K-means, hierarchical clustering) to group patients based on cognitive severity.
- **Regression models** (e.g., linear regression) to predict cognitive decline based on initial conditions.
- **ANOVA (Analysis of Variance)** to determine if significant differences exist between the cognitive impairment levels across different groups of patients.

IV. RESULTS

The study reveals that the severity of COVID-19 has a significant impact on cognitive performance and fatigue. Patients who had a severe form of the disease show significantly lower memory scores ($p < 0.001$), longer reaction times ($p < 0.001$), and higher fatigue scores ($p < 0.001$) compared to the Mild and Moderate groups. However, age has no significant effect on these variables ($p > 0.05$).

Principal Component Analysis (PCA) shows that the first component (PC1), explaining 36.66% of the variance, is positively influenced by memory and negatively by fatigue and reaction time. The second component (PC2), explaining 22.40% of the variance, is correlated with age and COVID-19 severity.

Clustering identifies three distinct patient profiles, differentiated by COVID-19 severity, gender, cognitive performance, and fatigue, highlighting the heterogeneity of post-COVID sequelae.

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V. DISCUSSION

Our findings suggest a correlation between COVID-19 severity and long-term cognitive deficits. Patients with severe symptoms tend to have slower reaction times and higher

fatigue scores. The regression analysis supports this, showing that higher initial symptom scores predict worse long-term outcomes. The clustering approach effectively distinguishes patient groups, which may aid in targeted interventions. ANOVA results further corroborate these findings, with significant cognitive differences across severity levels. However, limitations such as sample bias and missing data require further exploration.

VI. FUTURE WORK

Future research should consider expanding the dataset with additional neuroimaging biomarkers and employing deep learning models to improve predictive accuracy. Exploring genetic predisposition factors related to cognitive decline post-COVID-19 is another promising direction.

VII. CONCLUSION

This study confirms that COVID-19 severity significantly impacts cognitive performance and fatigue. Patients with severe cases show lower memory scores, longer reaction times, and higher fatigue levels. While age has no significant effect, PCA and clustering reveal distinct cognitive profiles based on severity, sex, and fatigue. These findings highlight the need for further research into long-term neurological consequences of COVID-19.